Point convergence at the printed reflected-gradient cap and explicit composite rates

Abstract

We prove point convergence of the exact printed-cap adaptive reflected proximal-gradient algorithm for arbitrary proper closed convex penalties. Initial points may lie outside the penalty domain. A finite-block zero-loss argument resolves the endpoint degeneracy without a stepsize upper bound. Under strong convexity and global smoothness we give explicit two-step geometric rates.

Keywords: reflected proximal gradient; adaptive step sizes; endpoint convergence; convex composite optimization; geometric rates

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1 Source, scope, and exact iteration

We study Algorithm 3 of Ou, Themelis and Latafat [1], at its printed curvature cap $c = \sqrt{6}/10$. Their convergence result requires a strict reduction of that cap. The theorem below concerns convex composite minimization and does not assert a nongradient variational-inequality result or a sharp dynamical threshold. The source is specifically arXiv:2609.15936v1, Algorithm 3, printed page 9, Lemma 2.3 and its appendix proof. The zero-vector convention below completes a quotient left unspecified in the printed algorithm.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and continuously differentiable, with locally Lipschitz gradient $F = \nabla f$. Let $g : \mathbb{R}^n \to (-\infty, +\infty]$ be proper, closed and convex, and suppose $S = \operatorname{argmin}(f + g) \neq \varnothing$. Choose arbitrary $x_0 \in \mathbb{R}^n$, $\gamma_0, L_0 > 0$, and set $x_{-2} = x_{-1} = x_0$, $\rho_0 = 1$. Write $h_k = F(x_k) - F(x_{k-1})$ and

$$a_k = \begin{cases} \dfrac{\max\{0, -\langle h_k, h_{k-1}\rangle\}}{\|h_k\|\|h_{k-1}\|}, & \|h_k\|\|h_{k-1}\| > 0, \\ 0, & \text{otherwise,} \end{cases} \qquad \beta_k = \sqrt{\frac{\sqrt{3}}{\sqrt{3} - 1 + a_k}}.$$

For $k \geq 1$ put $L_k = \|h_k\|/\|x_k - x_{k-1}\|$, with $0/0 = 0$; retain the supplied L_0 at index zero. With $c/0 = +\infty$, iterate

$$\begin{aligned} \gamma_{k+1} &= \min\{\gamma_k\sqrt{1/2 + \rho_k}, \gamma_k\beta_k, c/L_k\}, \quad \rho_{k+1} = \gamma_{k+1}/\gamma_k, \\ x_{k+1} &= \operatorname*{prox}_{\gamma_{k+1}g}\big(x_k - \gamma_{k+1}(F(x_k) + 2\rho_{k+1}h_k)\big). \end{aligned} \tag{1}$$

Every tie in the minimum is permitted. No stopping rule is imposed.

Theorem 1. *The iteration is well defined and converges to some $x_\infty \in S$. The stepsizes have a uniform positive lower bound. For $d_k = x_k - x_{k-1}$ and the selected subgradient*

$$p_k = -d_k/\gamma_k - F(x_{k-1}) - 2\rho_k h_{k-1} \in \partial g(x_k), \qquad k \geq 1,$$

one has $d_k \to 0$, $(f+g)(x_k) \to \min(f+g)$, and

$$p_k + F(x_k) \to 0, \quad \gamma_{k+1}(p_k + F(x_k)) \to 0, \quad \gamma_k(p_k + F(x_{k-1})) \to 0.$$

For each $z \in S$, the finite gap $G_k(z) = g(x_k) - g(z) + \langle F(x_k), x_k - z\rangle$ satisfies $G_k(z) \to 0$ and $\gamma_{k+1}G_k(z) \to 0$. These conclusions allow $x_0 \notin \operatorname{dom} g$ and require no upper bound on γ_k.

2 An exact-cap energy inequality

Fix $z \in S$. Proximal outputs are in $\operatorname{dom} \partial g$, so all quantities below are finite starting at index two. Convexity gives $G_k(z) \geq (f+g)(x_k) - (f+g)(z) \geq 0$. Define

$$U_k = \tfrac{1}{2}\|x_k - z\|^2 + \tfrac{1}{2}\|d_k\|^2 + 10(\gamma_k\rho_k)^2\|h_{k-1}\|^2 + (1+2\rho_k)\gamma_k G_{k-1}(z), \quad k \geq 2.$$

For a transition $k \geq 2$ abbreviate $s = \gamma_k$, $t = \gamma_{k+1}$, $q = \rho_k$, $r = t/s$, $a = a_k$, $w = th_k$, $v = tqh_{k-1}$ and $\tau = tL_k$. Put

$$H = 10r^2 + (1+2r)^2(1+2a/3), \qquad A = 1/2 + q - r^2,$$
$$B = 5/2 - r^2(1+3a/2), \qquad D = 1/2 + r^2 - \tau^2 H,$$
$$E = (2a/3)(1+2r)^2\|w\|^2 + 6a\|v\|^2 + 4(1+2r)\langle v, w\rangle.$$

Lemma 1. *For every such transition,*

$$U_k - U_{k+1} \geq 2sAG_{k-1} + 4(sq)^2 B\|h_{k-1}\|^2 + D\|d_k\|^2 + 2r(1+2r)\langle d_k, w\rangle + E. \tag{2}$$

All terms on the right are nonnegative. Moreover,

$$\|d_{k+1}\|^2 \leq \langle d_{k+1}, rd_k + 2v - (1+2r)w\rangle. \tag{P}$$

Proof. The proximal variational inequality is $\langle x_{k+1} - y, x_k - x_{k+1} - t(F(x_k) + 2rh_k)\rangle \geq t(g(x_{k+1}) - g(y))$ for $y \in \operatorname{dom} g$. Combining it at $y = z, x_k$ with the preceding proximal inequality at $y = x_{k+1}$ gives, by expanding squared distances (equivalently the algebra of source Lemma 2.3),

$$N_{k+1} \leq \tfrac{1}{2}\|x_k - z\|^2 + 2trG_{k-1} + 4\|v\|^2 - r^2\|d_k\|^2 + (1+2r)^2\|w\|^2 - 2r(1+2r)\langle d_k, w\rangle - 4(1+2r)\langle v, w\rangle,$$

where $N_{k+1} = U_{k+1} - 10(tr)^2\|h_k\|^2$. Subtracting from U_k and retaining the displayed cross term gives (2) exactly after adding and subtracting the first two terms defining E. If $\langle v, w\rangle < 0$, the definition of a gives $E = a(\sqrt{2/3}(1+2r)\|w\| - \sqrt{6}\|v\|)^2$. If $\langle v, w\rangle \geq 0$, $a = 0$ and $E = 4(1+2r)\langle v, w\rangle \geq 0$, including zero vectors.

Let $b = \sqrt{3}$ and $R = (1+\sqrt{3})/2$, so $R^2 = 1/2 + R$. The growth branch implies $A \geq 0$ and $0 < \rho_k \leq R$. The second branch gives $r^2 \leq b/(b-1+a)$ and hence

$$B \geq \frac{(1-a)(3b/2 - 5/2)}{b-1+a} \geq 0.$$

Its only zero on the closed admissible set is $(r, a) = (1, 1)$. Direct expansion, using $c^2 = 3/50$, gives

$$D = Q(r,a)/25 + (c^2 - \tau^2)H, \qquad Q = 10(1-r) + (1-a)(1+2r)^2.$$

For $r \leq 1$, $Q \geq 0$, with equality only at $r = a = 1$. For $r > 1$, $1 - a \geq b(1 - r^{-2})$, so

$$Q \geq (r-1)\{-10 + b(r+1)(2+1/r)^2\} > 0,$$

since $(r+1)(2+1/r)^2 > 8$ and $8\sqrt{3} > 10$. Therefore $D \geq 0$, and $D = 0$ precisely when $(r, a, \tau) = (1, 1, c)$. Monotonicity of F handles the remaining inner product. Finally, insert the two actual selected subgradients into $\langle d_{k+1}, p_{k+1} - p_k\rangle \geq 0$ and multiply by t to obtain (P). This is an inequality, not the identity valid when $g = 0$. ∎

3 Endpoint point convergence

Proof of Theorem 1. The energy decreases from finite U_2. It bounds the orbit, all d_k, all w_k (because $\|w\| \leq c\|d_k\|$), and all v_k (because $r \leq R$). Each nonnegative decrement in (2) tends to zero. In particular

$$D\|d_k\|^2 \to 0, \quad E_k \to 0, \quad 2r(1+2r)\langle d_k, w\rangle \to 0, \quad 4(sq)^2 B\|h_{k-1}\|^2 \to 0.$$

Suppose a subsequence has $d_{k_j} \to d \neq 0$. Extract limits of all bounded parameters and scaled vectors at this and the next index. The unique zero of D forces

$$r \to 1,\ a \to 1,\ \tau \to c, \quad w \to w_*, \quad \|w_*\| = c\|d\| > 0, \quad \langle d, w_*\rangle = 0, \quad v \to -w_*.$$

The last claim follows from the limiting identity $E = 6\|v + w_*\|^2$. Let $d_{k_j+1} \to e$ and use primes for next-transition quantities. Exactly $v' = r'rw$. If $e = 0$, then $w' \to 0$, while the next history decrement is $4B'\|rw\|^2 \to 0$. It forces $B' \to 0$, hence $r', a' \to 1$, and then $v' \to w_*$. The next E would tend to $6\|w_*\|^2 > 0$, a contradiction. Thus $e \neq 0$. Applying the same zero-loss conclusions again gives $w' \to -w_*$, $\langle e, -w_*\rangle = 0$, and $\|e\| = \|d\|$. Passing to (P),

$$\|e\|^2 \leq \langle e, d - 5w_*\rangle = \langle e, d\rangle \leq \|e\|\|d\| = \|e\|^2,$$

so $e = d$.

If increments did not vanish, choose such a subsequence with $\|d\| \geq \epsilon > 0$. Let Δ be the bounded orbit's diameter and choose a fixed integer N with $N\epsilon > \Delta$. A single subsequence can be extracted for offsets $0, \ldots, N$. Repeated application of the adjacent-index argument gives $d_{k_j+i} \to d$, so $x_{k_j+N} - x_{k_j} \to Nd$, contradicting the diameter. Hence $d_k \to 0$.

Local Lipschitz continuity on a compact convex neighborhood of the bounded orbit gives $L_k \leq M < \infty$ for $k \geq 1$. Put $\sigma = \min(c/L_0, c/M)$ and $m = \min(\gamma_0, \sigma)/\sqrt{2} > 0$. Every cap is at least σ, every beta multiplier is at least one, and every growth multiplier is at least $1/\sqrt{2}$. If $\gamma_{k+1} < m$ is the first violation, only growth can produce it, implying $\rho_k < 1/2$ and $\gamma_k < \sqrt{2}m \leq \sigma$. Index zero is impossible. At $k \geq 1$, the preceding cap could not produce $\gamma_k < \sigma$, so $\rho_k \geq 1/\sqrt{2}$, a contradiction. Thus all $\gamma_k \geq m$.

The preceding cap gives $\gamma_k \|h_{k-1}\| \leq c\|d_{k-1}\| \to 0$. Therefore

$$\gamma_k(p_k + F(x_{k-1})) = -d_k - 2\gamma_k\rho_k h_{k-1} \to 0.$$

Divide by the positive lower bound and use $h_k \to 0$ to obtain $p_k + F(x_k) \to 0$. Importantly, the next-step weighted identity is

$$\gamma_{k+1}(p_k + F(x_k)) = -rd_k + w - 2v \to 0.$$

No upper stepsize bound was used. Every cluster point x_* satisfies $-F(x_*) \in \partial g(x_*)$ by the closed graph of the convex subdifferential; thus it belongs to S. This also establishes domain membership without continuity of g.

The subgradient inequality gives $0 \leq G_k(z) \leq \langle p_k + F(x_k), x_k - z\rangle$. Boundedness and the two residual limits give $G_k(z) \to 0$ and $\gamma_{k+1}G_k(z) \to 0$. All nondistance terms of U_k vanish. Thus $\|x_k - z\|$ has a limit for every $z \in S$. Take a cluster point in S; its distance vanishes along a subsequence and consequently along the entire sequence. Objective convergence follows from the gap bound. ∎

4 Explicit strong-composite geometric rates

Assume now that f is globally μ-strongly convex and ∇f is globally M-Lipschitz, where $\mu > 0$. An affine minorant for proper closed convex g makes $f + g$ coercive; it has a unique minimizer z. Retain the energy and set

$$R = (1 + \sqrt{3})/2, \quad m = \frac{1}{\sqrt{2}} \min\{\gamma_0, c/L_0, c/M\}, \quad \eta = \mu m,$$
$$\alpha = R + c, \quad \beta = 2R^2 c, \quad P = R + (1 + 2R)c, \quad \delta = \min\{1/5, 2\eta\},$$
$$C = \tfrac{1}{2}\{(\alpha/\eta + P)^2 + (\beta/\eta + \beta)^2\} + \tfrac{1}{2}(P^2 + \beta^2)$$
$$+ 10R^2c^2 + \frac{1 + 2R}{\eta}(\alpha^2 + \beta^2), \qquad \theta = \frac{C}{C + \delta}.$$

Theorem 2. *With $r_k = p_k + F(x_k)$, the following bounds hold:*

$$\begin{aligned} U_k &\le U_2\theta^{\lfloor (k-2)/2\rfloor}, && k \ge 2,\\ \|x_k - z\| &\le \sqrt{2U_2}\,\theta^{\lfloor (k-2)/2\rfloor/2}, && k \ge 2,\\ 0 \le (f+g)(x_k) - (f+g)(z) \le G_k(z) &\le (U_2/m)\theta^{\lfloor (k-1)/2\rfloor}, && k \ge 1,\\ \|r_k\|^2 &\le \frac{4(\alpha^2+\beta^2)U_2}{m^2}\theta^{\lfloor (k-3)/2\rfloor}, && k \ge 3. \end{aligned}$$

The constants are explicit and conservative; no optimality is claimed.

Proof. Strong monotonicity gives $\langle d_k, h_k\rangle \ge \mu\|d_k\|^2$ and $\|x_k - z\| \le \|r_k\|/\mu$. In (2), if $r \le 1/2$, then $Q \ge 5$ and $D \ge 1/5$. If $r \ge 1/2$, its monotonicity term is at least $2\mu m\|d_k\|^2$. Hence $U_k - U_{k+1} \ge \delta\|d_k\|^2$ for every $k \ge 2$.

Put $X = \|d_k\|$, $Y = \|d_{k-1}\|$. The present and preceding caps imply $\|w\| \le cX$ and $\|v\| \le R^2cY$. The weighted residual and (P) imply, with $Z = \alpha X + \beta Y$ and $E_0 = PX + \beta Y$,

$$t\|r_k\| \le Z, \quad \|x_k - z\| \le Z/\eta, \quad tG_k \le Z^2/\eta, \quad \|d_{k+1}\| \le E_0.$$

Therefore

$$U_{k+1} \le \tfrac{1}{2}(Z/\eta + E_0)^2 + \tfrac{1}{2}E_0^2 + 10R^2c^2X^2 + (1+2R)Z^2/\eta \le C(X^2 + Y^2).$$

The last inequality applies $(aX + bY)^2 \le (a^2 + b^2)(X^2 + Y^2)$ separately to the three squares. Summing the decrement at $k-1, k$, for $k \ge 3$, yields $U_{k+1} \le \theta U_{k-1}$. Since $U_3 \le U_2$, induction on both parities proves the energy bound. Its distance term gives the second bound; $U_{k+1} \ge mG_k$ gives the third, including $k = 1$. Finally

$$\|r_k\|^2 \le (\alpha^2+\beta^2)(\|d_k\|^2 + \|d_{k-1}\|^2)/m^2 \le 4(\alpha^2+\beta^2)U_{k-1}/m^2.$$

This proves the last bound. Zero increments require no division. At a stationary minimizer, future stepsizes may grow without bound, without affecting any estimate. ∎

5 Version and reproducibility

The pinned primary PDF has SHA-256

`5596cbb161bfc0fff3347636c768d19eebfb`
`6d9e1a0511fd51ef06193187bb1e`.

The accompanying symbolic scripts check coefficient decompositions and rate constants. The infinite-time assertions are proved analytically above; those computations are auxiliary algebra checks. The conclusions concern precisely the completed algorithm (1), rather than every possible zero-cosine convention or any nongradient operator.

References

[1] H. Ou, A. Themelis and P. Latafat. *An Adaptive Linesearch-free Method for Monotone Variational Inequalities under Local Lipschitz Continuity*. arXiv:2609.15936v1, 2026. `https://arxiv.org/abs/2609.15936v1`.